Day 14 — Common Network Ports

1. Introduction

Network ports are logical communication endpoints that allow computers to identify which application or service should receive network traffic. Each port is identified by a number ranging from 0 to 65535 and works with transport protocols such as TCP or UDP.

Understanding common network ports is essential in computer networking and cybersecurity because it helps security professionals identify network services, analyze traffic, configure firewalls, and detect potentially unauthorized access.

2. Common Network Ports Reference Sheet

You can copy this table into Google Sheets and use it as your deliverable.

Port	Protocol	Transport	Purpose
20	FTP Data	TCP	Transfers file data in active FTP
21	FTP	TCP	Transfers files and manages FTP sessions
22	SSH	TCP	Secure remote login and command execution
23	Telnet	TCP	Remote terminal access without encryption
25	SMTP	TCP	Transfers email between mail servers
53	DNS	TCP/UDP	Resolves domain names to IP addresses
67	DHCP Server	UDP	Provides IP configuration to clients
68	DHCP Client	UDP	Receives IP configuration from a DHCP server
80	HTTP	TCP	Transfers web content without transport encryption
110	POP3	TCP	Retrieves email from a mail server
123	NTP	UDP	Synchronizes system clocks
143	IMAP	TCP	Accesses and manages email on a server
161	SNMP	UDP	Monitors and manages network devices
389	LDAP	TCP/UDP	Accesses directory services

443	HTTPS	TCP	Secure web communication, commonly using TLS
445	SMB	TCP	File and printer sharing on Windows networks
465	SMTPS	TCP	Email submission using implicit TLS
587	SMTP Submission	TCP	Sends outgoing email, commonly with STARTTLS
993	IMAPS	TCP	Secure email access using IMAP
995	POP3S	TCP	Secure email retrieval using POP3
3306	MySQL	TCP	Connects to MySQL databases
3389	RDP	TCP/UDP	Remote Desktop access
5432	PostgreSQL	TCP	Connects to PostgreSQL databases
8080	HTTP Alternate	TCP	Common alternate web or application port

Note: These are common default ports, not guarantees that a service is running on that port. Administrators can configure services to use different ports.

3. Explanation of Important Network Services

HTTP — Port 80

HTTP stands for Hypertext Transfer Protocol. It allows web browsers and web servers to exchange web pages and other resources. Standard HTTP does not encrypt the communication, so sensitive information should not be transmitted through an unprotected HTTP connection.

HTTPS — Port 443

HTTPS stands for Hypertext Transfer Protocol Secure. It uses TLS to protect data exchanged between a browser and a web server. It helps provide confidentiality, integrity, and server authentication when correctly configured.

DNS — Port 53

DNS stands for Domain Name System. It translates human-readable domain names, such as example.com, into IP addresses. DNS commonly uses UDP port 53 for ordinary queries, while TCP port 53 is also used for certain responses and other DNS operations.

SSH — Port 22

SSH stands for Secure Shell. It enables encrypted remote access to computers and servers. It is commonly used by system administrators and cybersecurity professionals to execute commands, manage systems, and transfer files securely.

SMTP — Port 25

SMTP stands for Simple Mail Transfer Protocol. It is primarily used to send and relay email between mail servers. Email clients commonly submit outgoing messages through ports 587 or 465, depending on the configuration.

FTP — Ports 20 and 21

FTP stands for File Transfer Protocol. Port 21 is used for control communication, while port 20 is traditionally used for data transfer in active FTP. Standard FTP does not encrypt credentials or transferred data, making secure alternatives such as SFTP preferable for sensitive files.

4. Why Network Ports Matter in Cybersecurity

Network ports help security professionals understand which services are exposed on a system. An unnecessarily open port can provide an attacker with an opportunity to probe for vulnerabilities or attempt unauthorized access.

Security teams use port information during network scanning, firewall configuration, penetration testing, incident investigation, and system hardening. For example, SSH on port 22 should be restricted to authorized users, and database ports such as 3306 should not generally be exposed to the public internet unless there is a carefully controlled requirement.

Port numbers alone do not establish that a system is vulnerable. Security assessments must also consider the service version, configuration, authentication, network exposure, and applicable security controls.